SC Honors Geometry - Nappier
Chapter 5 Review • Congruent Triangles

Name: ____________________________________

5.1 Angles of triangles.

1. Two angles of a triangle measure 52° and 71°. Find the third angle.

2. The angles of a triangle measure $(4x)°$, $(3x + 6)°$, and $(2x - 6)°$. Find x and all three angles.

3. An exterior angle measures 138° and one remote interior angle measures 61°. Find the other remote interior angle.

4. The acute angles of a right triangle measure $(6x - 3)°$ and $(3x + 3)°$. Find x and both angles.

5. Classify a triangle with sides 8, 8, 8 and angles 60°, 60°, 60°.

5.2 Congruent polygons.

6. $\triangle ABC \cong \triangle DEF$. Name the side that corresponds to $\overline{AB}$ and the angle that corresponds to $\angle C$.

7. $\triangle ABC \cong \triangle DEF$, $BC = 5x - 2$, and $EF = 23$. Find x.

8. $\triangle ABC \cong \triangle DEF$, $m\angle A = (2y + 14)°$, and $m\angle D = 48°$. Find y.

9. Name the property: if $\triangle PQR \cong \triangle STU$, then $\triangle STU \cong \triangle PQR$.

10. A classmate writes $\triangle ABC \cong \triangle FDE$ after being told $\triangle ABC \cong \triangle DEF$. Explain the error.

Triangle rigidity.

11. Name the criterion: two pairs of congruent sides and the pair of congruent angles between them.

12. Name the criterion: two pairs of congruent angles and a pair of congruent sides NOT between them.

13. Two right triangles have congruent hypotenuses and one pair of congruent legs. Name the criterion.

14. Explain why SSA is not a criterion.

15. $\triangle MNO \cong \triangle PQR$, $MN = 6x - 7$, and $PQ = 29$. Find x.

5.4 Equilateral and isosceles triangles.

16. The vertex angle of an isosceles triangle measures 38°. Find each base angle.

17. The base angles of a triangle measure $(2x + 21)°$ and $(5x - 9)°$. Find x and each base angle.

18. The legs of an isosceles triangle measure $7x - 12$ and $4x + 9$. Find x and the length of each leg.

5.7 Using congruent triangles.

19. What does CPCTC stand for, and at what point in a proof may you use it?

20. A surveyor marks C so that $\overline{AC} \cong \overline{CD}$ and $\overline{BC} \cong \overline{CE}$, then measures ED = 68 m. Find AB and justify.